Lecture 16

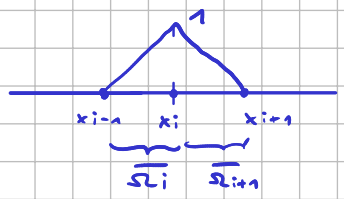
04.12.25

Example (Linear finite elements in 1D) Consider $-u''(x) = f(x)$ in $\Omega = (0,1)$, $u(0) = u(1) = 1$.

Admissible decomposition: $\bar{\Omega} = \bigcup_{i=1}^{n_e} \bar{\Omega}_i$, $\Omega_i = (x_{i-1}, x_i)$, $0 = x_0 < x_1 < \dots < x_N = x_{N+1} = 1$.

Let us set $h_i := x_i - x_{i-1}$, $i = 1, \dots, N+1$, $n_e = N+1$ and define a basis of linear finite element functions

$$\varphi_i(x) = \begin{cases} \frac{x - x_{i-1}}{h_i} & \text{if } x \in \bar{\Omega}_i \\ \frac{x_{i+1} - x}{h_{i+1}} & \text{if } x \in \bar{\Omega}_{i+1} \\ 0 & \text{otherwise} \end{cases}$$



Note: φ_i is piecewise linear in Ω with $\varphi_i(x_j) = \delta_{ij}$, moreover, $\varphi_i \in H^1(\Omega)$ and $\varphi_i \in C^0(\bar{\Omega})$

We then use $V_n = \text{span}\{\varphi_i \mid i = 1, \dots, n = N\}$ and note that

$$\varphi_i'(x) = \begin{cases} \frac{1}{h_i} & \text{if } x \in \Omega_i \\ -\frac{1}{h_{i+1}} & \text{if } x \in \Omega_{i+1} \\ 0 & \text{else} \end{cases}$$

$\Rightarrow$ for the Galerkin matrix $A_n = [a(\varphi_i, \varphi_j)]_{i,j=1}^{n}$ we explicitly compute

$$a(\varphi_i, \varphi_j) = \int_\Omega \varphi_i'(x)\varphi_j'(x)\,dx = \begin{cases} \int_{\Omega_i \cup \Omega_{i+1}} (\varphi_i'(x))^2\,dx & \text{if } i = j \\ \int_{\Omega_{i+1}} \varphi_i'(x)\varphi_{i+1}'(x)\,dx & \text{if } j = i+1 \\ \int_{\Omega_i} \varphi_i'(x)\varphi_{i-1}'(x)\,dx & \text{if } j = i-1 \\ 0 & \text{else} \end{cases}$$

$$= \begin{cases} h_i \cdot \frac{1}{h_i^2} + h_{i+1} \cdot \frac{1}{h_{i+1}^2} & \text{if } i = j \\ -\frac{1}{h_{i+1}} \frac{1}{h_{i+1}} h_{i+1} & \text{if } j = i+1 \\ -\frac{1}{h_i} \frac{1}{h_i} h_i & \text{if } j = i-1 \\ 0 & \text{else} \end{cases} = \begin{cases} \frac{1}{h_i} + \frac{1}{h_{i+1}} & \text{if } i = j \\ -\frac{1}{h_{i+1}} & \text{if } j = i+1 \\ -\frac{1}{h_i} & \text{if } j = i-1 \\ 0 & \text{else} \end{cases}$$

We then obtain the following matrix:

$$A_n = \begin{bmatrix} \frac{1}{h_1} + \frac{1}{h_2} & -\frac{1}{h_2} & & \\ -\frac{1}{h_2} & \frac{1}{h_2} + \frac{1}{h_3} & \ddots & \\ & \ddots & \ddots & -\frac{1}{h_n} \\ & & -\frac{1}{h_n} & \frac{1}{h_n} + \frac{1}{h_{n+1}} \end{bmatrix} \in \mathbb{R}^{n \times n}$$

$$\xRightarrow[\text{uniform mesh}]{\text{if } h = h_i} \underbrace{\frac{1}{h} \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{bmatrix}}_{\text{looks familiar?!}} \in \mathbb{R}^{n \times n}$$

For the right hand side we generally obtain:

$$f_n = \begin{bmatrix} F(\varphi_1) \\ \vdots \\ F(\varphi_n) \end{bmatrix} = \begin{bmatrix} \int_\Omega f(x)\varphi_1(x)\,dx \\ \vdots \\ \int_\Omega f(x)\varphi_n(x)\,dx \end{bmatrix}$$

$\rightsquigarrow$ typically approximated by numerically quadrature, i.e.

$$\int_\Omega f(x)\varphi_i(x)\,dx \approx \sum_j \omega_{ji} f(x_j)\varphi_i(x_j)$$

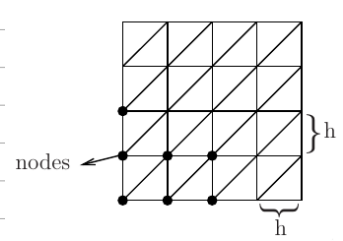
Example (Linear finite elements in 2D) Consider $-\Delta u = f$ in $\Omega = (0,1)\times(0,1)$

$u = 0$ on $\partial\Omega$

Step 1 Decomposition of Ω as follows

a) Decomposition of $\bar{\Omega}$ into squares with edge length $h = \frac{1}{N+1}$, $N \in \mathbb{N}$ $\Rightarrow$ $(N+1)^2$ squares

b) Cut each square along one of the diagonals into two triangles $\Rightarrow$ $n_e = 2(N+1)^2$ elements

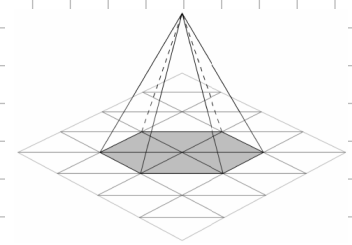


$\rightarrow$ nodes $x_{\ell m} = (\ell h, mh)$, $\ell, m = 0, \dots, N+1$ $\Rightarrow$ $(N+2)^2$ nodes, N^2 inner nodes

For homogeneous Dirichlet b.c.: inner nodes $\hat{=}$ dimension of FEM space

$$V_n = \text{span}\left\{ \varphi_{\ell m} \in C^0(\bar{\Omega}) \,\middle|\, \varphi_{\ell m}(x_{\tilde{\ell}\tilde{m}}) = \begin{cases} 1 & \text{if } \tilde{\ell} = \ell, \tilde{m} = m \\ 0 & \text{else} \end{cases}, \varphi_{\ell m} \text{ piecewise linear} \right\}$$

$\rightarrow$ basis functions are called pyramid/tent/hat functions and on each element read



$$\varphi_{\ell m}(x_1, x_2) = a_0^{\ell m} + a_1^{\ell m} x_1 + a_2^{\ell m} x_2 \quad \text{for all } (x_1, x_2) \in \Omega_i$$

Example (more general Lagrange finite elements in 2D)

Recall from numerics 1: given a set of grid values/points (x_i, y_i), $i = 1, \dots, d$, $x_i \neq x_j$, $i \neq j$

$\Rightarrow$ there exists a unique polynomial p of deg. $d-1$ that interpolates, i.e., $p(x_i) = y_i$, $i = 1, \dots, d$

$\Rightarrow$ we even have the explicit formula for p: $p(x) = \sum_{i=1}^{d} y_i \, \ell_i(x)$, with Lagrange polynomial

$$\ell_i(x) = \prod_{\substack{0 \le m \le d \\ m \neq i}} \frac{x - x_m}{x_i - x_m} \quad \Rightarrow \quad \ell_i(x_j) = \delta_{ij}$$

$\rightarrow$ can also be done in 2D!

Common FEM choices

- linear finite elements on a triangle 3 nodes

$$\varphi_i|_{\Omega_j}(x_1, x_2) = a_0^j + a_1^j x_1 + a_2^j x_2$$

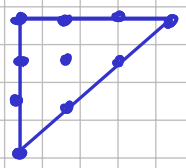
degree of freedom of one element: 3

- quadratic finite elements on a triangle 6 nodes

$$\varphi_i|_{\Omega_j}(x_1, x_2) = a_0^j + a_1^j x_1 + a_2^j x_2 + a_3^j x_1 x_2 + a_4^j x_1^2 + a_5^j x_2^2$$

degree of freedom of one element: 6

- cubic finite elements on a triangle

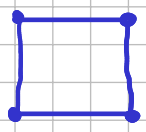


10 nodes

$$\varphi_i|_{\Omega_j}(x_1,x_2) = a_0^j + a_1^j x_1 + \ldots + a_5^j x_2^2 + a_6^j x_1^3 + a_7^j x_1^2 x_2 + a_8^j x_1 x_2^2 + a_9^j x_2^3$$

degree of freedom of one element: 10

- bilinear finite elements on a quatrilateral

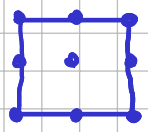


4 nodes

$$\varphi_i|_{\Omega_j}(x_1,x_2) = a_0^j + a_1^j x_1 + a_2^j x_2 + a_3^j x_1 x_2$$

degree of freedom of one element: 4

- biquadratic finite elements on a quatrilateral



9 nodes

$$\varphi_i|_{\Omega_j}(x_1,x_2) = a_0^j + a_1^j x_1 + a_2^j x_2 + a_3^j x_1 x_2 + a_4^j x_1^2 + a_5^j x_2^2 + a_6^j x_1^2 x_2 + a_7^j x_1 x_2^2 + a_8^j x_1^2 x_2^2$$

degree of freedom of one element: 9

<u>Important note</u>: the dimension of the resulting finite-dimensional subspace V_n composed of such elements is **not** simply given by dof (element) $\times$ n_e but depends on the overall nodes of the mesh as well as on the boundary conditions!